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Quick review of binomial (and multinomial) coefficients

The Binomial Coefficient $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ has several (easy) interpretations

= # words with k 1's
($n-k$) 2's,

e.g. $\binom{4}{2} = 6 \rightarrow$
= # $\{ \begin{matrix} 1122, 1212, 1221, \\ 2112, 2121, 2211 \end{matrix} \}$ i.e., rearrangements of $\underbrace{11\dots 1}_k \underbrace{22\dots 2}_{n-k}$

= # lattice paths in $\mathbb{Z}^2$ taking E east or N north steps, from $(0,0)$ to $(k, n-k)$



$\leftrightarrow$ $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ N & E & N & E & E & E & N & E & E \end{matrix} \leftarrow \binom{9}{3}$
(could be 1's/2's instead of E's/N's)

= # subsets of $[n] := \{1, 2, \dots, n\}$ of size k

e.g. $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ N & E & E & N & E & E & E & N & E \end{matrix} \leftrightarrow \{2, 3, 5, 6, 7, 9\} \subseteq [9]$
(position of E's in word)

Of course, we have

Thm (Binomial Theorem) $(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k}$

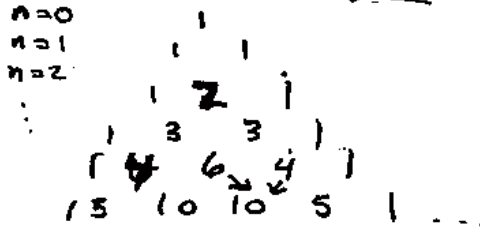
Lemma (Pascal's Identity)

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}$$

Pf (bijection):

$A \subseteq [n] \mapsto \begin{cases} A & \text{if } n \notin A \\ A - \{n\} & \text{if } n \in A \end{cases}$

Pascal's triangle



Multinomials

How many rearrangements (anagrams) of BANANAS?

i.e., of 3A's, 1B, 2N's, 1S? (equiv. of $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ A & A & A & B & N & N & S \end{matrix}$)

∃ a transitive action of the symmetric group G_7^V of permutations of $[7]$ on the rearrangements

e.g., perm. $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 1 & 4 & 3 & 6 & 7 & 5 \end{pmatrix}$ sends $\begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ A & A & A & B & N & N & S \end{matrix} \rightarrow \begin{matrix} 2 & 1 & 4 & 3 & 6 & 7 & 5 \\ A & A & B & A & N & S & N \end{matrix}$

The stabilizer of this action is $G_3 \times G_1 \times G_2 \times G_1 \subseteq G_7$
 $G_{\{1,2,3\}} \times G_{\{4\}} \times G_{\{5,6\}} \times G_{\{7\}}$

So by Orbit-Stabilizer Thm, # rearrangements = $\frac{\text{size of } G_7}{\text{size of stabilizer}} = \frac{7!}{3!1!2!1!} = \binom{7}{3,1,2,1}$

The Multinomial Coefficient $\binom{n}{k_1, k_2, \dots, k_m} = \frac{n!}{k_1! k_2! \dots k_m!}$ for $n = k_1 + k_2 + \dots + k_m$

= # words $w/$ k_1 1's, k_2 2's, ..., k_m m's, i.e., rearrang's of $\underbrace{1 \dots 1}_{k_1} \underbrace{2 \dots 2}_{k_2} \dots \underbrace{m \dots m}_{k_m}$

= # lattice paths in $\mathbb{Z}^m$ taking steps $\vec{e}_1, \vec{e}_2, \dots, \vec{e}_m$ from $O = (0, \dots, 0)$ to $(k_1, k_2, \dots, k_m)$
 (same correspondence between words + walks as w/ binomials)

= # chains $\emptyset = S_0 \subset S_{k_1} \subset S_{k_1+k_2} \subset \dots \subset S_{n=k_1+\dots+k_m} = [n]$ of subsets of $[n] = \{1, 2, \dots, n\}$ for which $\# S_i = i \quad \forall i = 0, k_1, k_1+k_2, \dots, n$

e.g. for $\binom{7}{3,1,2,1}$ have $2131314 \leftrightarrow \emptyset \subset \{2,4,6\} \subset \{1,2,4,6\} \subset \{1,3,4,5,6\} \subset [7]$

Note $\binom{n}{k} = \binom{n}{n-k}$ in multinomial notation

Also note: $\binom{n}{k_1, k_2, \dots, k_m} = \binom{n}{k_1} \cdot \binom{n-k_1}{k_2} \cdot \binom{n-k_1-k_2}{k_3} \cdot \dots \cdot \binom{n-k_1-\dots-k_{m-1}}{k_m}$

And... Multinomial Theorem: $(x_1 + x_2 + \dots + x_m)^n = \sum_{\substack{(k_1, k_2, \dots, k_m) \\ k_1+k_2+\dots+k_m=n}} \binom{n}{k_1, k_2, \dots, k_m} x_1^{k_1} x_2^{k_2} \dots x_m^{k_m}$

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Permutations and cycles (Stanley § 1.3)

Recall $S_n (= \mathcal{G}_n)$ = Symmetric group on n letters
 = permutations of $[n] := \{1, 2, \dots, n\}$
 = $\{\text{bijections } \sigma: [n] \rightarrow [n]\}$

Notations • two-line $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 \\ 7 & 1 & 3 & 12 & 2 & 10 & 5 & 4 & 11 & 6 & 9 & 8 \end{pmatrix}$

• one-line $\sigma = (7, 1, 3, 12, 2, 10, 5, 4, 11, 6, 9, 8)$

digraph =
directed graph

• functional digraph: $\sigma = \begin{matrix} 1 \rightarrow 7 \\ 7 \rightarrow 12 \\ 12 \rightarrow 8 \\ 8 \rightarrow 2 \\ 2 \rightarrow 10 \\ 10 \rightarrow 6 \\ 6 \rightarrow 9 \\ 9 \rightarrow 4 \\ 4 \rightarrow 11 \\ 11 \rightarrow 3 \\ 3 \rightarrow 5 \\ 5 \rightarrow 1 \end{matrix}$

• cycle notation: $\sigma = (1\ 7\ 5\ 2)\ (3)\ (4\ 12\ 8)\ (6\ 10)\ (9\ 11)$
 $= (8\ 4\ 12)\ (10\ 6)\ (5\ 2\ 7)\ (3)\ (11\ 9)$
 $= \text{etc.} = (3)\ (7\ 5\ 2\ 1)\ (10\ 6)\ (11\ 9)\ (12\ 8\ 4)$

cycle type of σ
 = partition of sizes of cycles of σ

standard form:
 • each cycle has its biggest element first
 • cycles appear w/ biggest elements increasing left-to-right

Q: How many $\sigma \in S_n$ of cycle type $\lambda = (\lambda_1, \lambda_2, \dots) \vdash n$
 $= 1^{c_1} 2^{c_2} 3^{c_3} \dots$

e.g. $n=4$

$\lambda = 1^4 = \begin{matrix} \square \\ \square \\ \square \\ \square \end{matrix} (a)(b)(c)(d) \quad 1$
 $2^1 1^2 = \begin{matrix} \square & \square \\ \square & \square \end{matrix} (ab)(c)(d) \quad \binom{4}{2} = 6$
 $2^2 = \begin{matrix} \square & \square \\ \square & \square \end{matrix} (ab)(cd) \quad \binom{4}{2}/2 = 6/2 = 3$
 $3^1 1^1 = \begin{matrix} \square & \square & \square \\ \square & \square & \square \end{matrix} (abc)(d) \quad 2! \cdot \binom{4}{3} = 2 \cdot 4 = 8$
 $4^1 = \begin{matrix} \square & \square & \square & \square \end{matrix} (abcd) \quad 3! \cdot \binom{4}{4} = 6$

multiplicity notation:

e.g. $\lambda = (5, 5, 5, 3, 2, 2, 2, 1, 1)$
 $= 1^2 2^4 3^1 4^0 5^3$
 i.e. $c_1=2, c_2=4, c_3=1, c_4=0, c_5=3$

Prop: There are $\frac{n!}{1^{c_1} c_1! 2^{c_2} c_2! 3^{c_3} c_3! \dots}$ perm's in S_n
 of cycle type $\lambda = 1^{c_1} 2^{c_2} 3^{c_3} \dots$

Proof of prop: Note that S_n acts on the set of perm's with cycle type $= \lambda$ transitively, by conjugation:

$$\text{e.g., } \underbrace{\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ a & b & c & d & e & f & g \end{pmatrix}}_{\pi} \cdot \underbrace{[(1234)(567)]}_{\sigma \text{ of type } 4'3'} \cdot \underbrace{\begin{pmatrix} a & b & c & d & e & f & g \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{pmatrix}}_{\pi^{-1}} = (abcd)(efg)$$

9/12 So the # of such perm's = size of the orbit
 $\xrightarrow{\text{orbit stabilizer}} = \frac{|S_n|}{|Z_{S_n}(\sigma_\lambda)|}$ if σ_λ is a perm. of type λ

where $Z_{S_n}(\sigma_\lambda) := \{\pi \in S_n : \pi \sigma_\lambda = \sigma_\lambda \pi\}$ is the centralizer of σ_λ in S_n .
 i.e. $\pi \sigma_\lambda \pi^{-1} = \sigma_\lambda$

Which perm's centralize $\sigma_\lambda = \underbrace{(a)(b) \dots}_{c_1 \text{ 1-cycles}} \underbrace{(cd)(ef) \dots}_{c_2 \text{ 2-cycles etc.}} \dots$?

- Products of powers of each cycle: there are $1^{c_1} 2^{c_2} 3^{c_3} \dots$ of these
- perm's that swap cycles of same size: there are $c_1! c_2! c_3! \dots$ of these (preserving cyclic order and biggest element)
- Products of those: $1^{c_1} c_1! 2^{c_2} c_2! \dots$ many

e.g. $\sigma_\lambda = (1234)(567)(8910)$

is centralized by $\pi = \underbrace{(4321)}_{(1234)^3} \underbrace{(59)(610)(78)}_{\text{swaps } (567) + (8910)}$

Thus $|\text{orbit}| = \frac{n!}{\prod_{j=1}^n j^{c_j} c_j!}$ as claimed. $\square$

Note: Stanley presents slightly different (but equivalent) proof by considering standard forms of perm's σ_λ .

There is an elegant reformulation of above in terms of g.f.'s:

Cor(Touchard) For $\sigma \in S_n$, let $c_k(\sigma) := \#$ of size k cycles of σ .

$$\sum_{n=0}^{\infty} \frac{1}{n!} \left(\sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} t_3^{c_3(\sigma)} \dots \right) x^n = e^{t_1 x + t_2 \frac{x^2}{2} + t_3 \frac{x^3}{3} + \dots}$$

$$= e^{\sum_{j=1}^{\infty} t_j \frac{x^j}{j}}$$

q/24 $(\mathcal{C}[t_1, t_2, t_3, \dots])[x]$

PS! (We'll see a more conceptual proof later...)

$$e^{t_1 x + t_2 \frac{x^2}{2} + \dots} = e^{t_1 x} e^{\frac{t_2 x^2}{2}} \dots$$

$$= \left(\sum_{c_1=0}^{\infty} \frac{(t_1 x)^{c_1}}{c_1!} \right) \left(\sum_{c_2=0}^{\infty} \frac{(t_2 \frac{x^2}{2})^{c_2}}{c_2!} \right) \dots$$

$$= \sum_{(c_1, c_2, \dots)} x^{1 \cdot c_1 + 2 \cdot c_2 + \dots} \frac{t_1^{c_1} t_2^{c_2} \dots}{1^{c_1} c_1! 2^{c_2} c_2! \dots}$$

$$= \sum_{n=0}^{\infty} x^n \frac{1}{n!} \sum_{\substack{(c_1, c_2, \dots) \\ \sum j \cdot c_j = n}} \left(\frac{n!}{1^{c_1} c_1! 2^{c_2} c_2! \dots} \right) t_1^{c_1} t_2^{c_2} \dots$$

$\hookrightarrow = \# \{ \sigma \in S_n : \begin{matrix} \text{ } \\ \sigma \text{ has } c_j \text{ } j\text{-cycles} \end{matrix} \}$
by previous prop.

$$= \sum_{n=0}^{\infty} x^n \frac{1}{n!} \sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} \dots$$

To a chord's theorem has many important consequences, a few of which we now review:

- ① DEF'N $c(n, k) := \# \{ \sigma \in S_n : \sigma \text{ has } k \text{ total cycles} \}$
(Signless) Stirling
numbers of 1st kind

I.e., $\sum_{k=1}^n c(n, k) t^k = \sum_{\sigma \in S_n} t^{\# \text{cycles}(\sigma)}$

Cor (to Touchard) $\sum_{k=1}^n c(n, k) t^k = t(t+1)(t+2) \dots (t+(n-1))$

9/27 Pf: Set $t_1 = t_2 = \dots = t$ in Touchard's thm to get

$$\begin{aligned} \sum_{n \geq 0} \frac{x^n}{n!} \underbrace{\sum_{\sigma \in S_n} t^{\# \text{cycles}(\sigma)}}_{\sum_{k=1}^n c(n, k) t^k} &= e^{t \left(\frac{x^1}{1} + \frac{x^2}{2} + \frac{x^3}{3} + \dots \right)} \\ &= e^{t(-\log(1-x))} \\ &= e^{\log((1-x)^{-t})} \\ &= (1-x)^{-t} \\ &= \sum_{n \geq 0} \binom{-t}{n} (-x)^n \\ &= \sum_{n \geq 0} \underbrace{\binom{t+n-1}{n}}_{= \frac{t(t+1) \dots (t+(n-1))}{n!}} x^n \end{aligned}$$

Pf follows by comparing coeffs of $x^n/n!$. $\square$

Remark: Prop The map $S_n \rightarrow S_n$
 $\sigma \mapsto \hat{\sigma} =$ put σ in standard cycle form and erase parentheses to view in-line not.

is a bijection, w/ $\# \text{cycles}(\sigma) = \# \text{left-to-right maxima in } \hat{\sigma}$.

Hence, $\sum_{\sigma \in S_n} t^{\# \text{L-to-R max}(\sigma)} = t(t+1) \dots (t+(n-1))$

e.g. $n=3$

σ	$\# \text{L-to-R maxima}$
<u>1</u> <u>2</u> <u>3</u>	3
<u>1</u> <u>3</u> <u>2</u>	2
<u>2</u> <u>1</u> <u>3</u>	2
<u>2</u> <u>3</u> <u>1</u>	2
<u>3</u> <u>1</u> <u>2</u>	1
<u>3</u> <u>2</u> <u>1</u>	1

Pf: (by example)

$\sigma \mapsto \hat{\sigma}$
 $(\underline{3})(\underline{7} \underline{5} \underline{2} \underline{1})(\underline{8} \underline{4} \underline{6}) \mapsto \underline{3} \underline{7} \underline{5} \underline{2} \underline{1} \underline{8} \underline{4} \underline{6}$
 is reversible. Just put $($ before each L-to-R maxima, and put $)$ right before the $($ and at the end. $\square$

(2) (Cor of Touchard) Can compute $E_k(n) :=$ expected # of k-cycles in a uniformly random $\sigma \in S_n$

$$E_k(n) = \frac{1}{n!} \sum_{\sigma \in S_n} C_k(\sigma) = \frac{1}{n!} \left[\frac{2}{k} \sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} \dots \right]_{t_1=t_2=\dots=1}$$

$$\begin{aligned} \text{So } \sum_{n \geq 0} E_k(n) x^n &= \left[\frac{2}{k} \sum_{n \geq 0} \frac{x^n}{n!} \sum_{\sigma \in S_n} t_1^{c_1(\sigma)} t_2^{c_2(\sigma)} \dots \right]_{t_1=t_2=\dots=1} \\ &= \left[\frac{2}{k} e^{t_1 \frac{x^1}{1} + t_2 \frac{x^2}{2} + t_3 \frac{x^3}{3} + \dots} \right]_{t_1=t_2=\dots=1} \\ &= \left[\frac{x^k}{k} e^{t_1 \frac{x^1}{1} + t_2 \frac{x^2}{2} + \dots} \right]_{t_i=1} \\ &= \frac{x^k}{k} e^{x^1 + \frac{x^2}{2} + \dots} = \frac{x^k}{k} e^{-\log(1-x)} = \frac{x^k}{(1-x)} \\ &= \sum_{n \geq k} \frac{1}{k} x^n \Rightarrow E_k(n) = \begin{cases} 1/k & \text{if } n \geq k, \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

Note: $E_k(n)$ eventually constant in n . In fact, one can show #k-cycles of random $\sigma \in S_n$ converges (as $n \rightarrow \infty$) to a Poisson random variable w/ expectation $\lambda = 1/k$.

(3) (Cor to Touchard) There are special classes of perms defined by restrictions on their cycle sizes, so all have nice g.f.'s.

e.g.: no large cycles

$\sigma \in S_n$ is an involution (i.e. $\sigma^2 = \text{identity}$) $\Leftrightarrow \sigma$ has only 1- and 2-cycles
 $(\sigma = (ab)(cd) \dots (xy) \dots)$

$$\begin{aligned} \text{So } \sum_{n \geq 0} \frac{x^n}{n!} \# \{ \text{involutions } \sigma \in S_n \} &= \left[e^{t_1 \frac{x^1}{1} + t_2 \frac{x^2}{2} + \dots} \right]_{t_1=t_2=\dots=1} \\ &= e^{x + \frac{x^2}{2}} \end{aligned}$$

and even $\sum_{n \geq 0} \frac{x^n}{n!} \sum_{\substack{\sigma \in S_n \\ \text{involution}}} t^{\# \text{ 1-cycles }(\sigma)} = e^{tx + \frac{x^2}{2}}$, etc.

What about no small cycles?

DEF 'N A derangement $\sigma \in S_n$ is a perm. w/ no fixed points,
i.e., w/ $\ell_1(\sigma) = 0$.

Q: (Derangement/
Hat-check problem): $n=100$ people check their hats;
attendant gives people's hats back randomly;
what is prob. that no one gets their ^{own} hat back?

a/29 i.e., what is $\frac{d_n}{n!}$, where $d_n := \# \{ \sigma \in S_n : \sigma \text{ is a derangement} \}$?

$$\begin{aligned} \sum_{n \geq 0} \frac{x^n}{n!} d_n &= \left[e^{t_1 \frac{x}{1} + t_2 \frac{x^2}{2} + \dots} \right]_{t_1=0, t_2=0, t_3=\dots=1} \\ &= e^{x^2/2 + x^3/3 + \dots} \\ &= e^{-\log(1-x) - \frac{x}{1}} = \boxed{\frac{e^{-x}}{1-x}} \end{aligned}$$

$$\begin{aligned} \text{But } e^{-x}/1-x &= (1+x+x^2+\dots)(1-\frac{x}{1!}+\frac{x^2}{2!}-\frac{x^3}{3!}+\dots) \\ &= \sum_{n \geq 0} x^n \left(1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \right) \end{aligned}$$

$$\begin{aligned} \text{So } \frac{d_n}{n!} &= 1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \dots + (-1)^n \frac{1}{n!} \\ &\xrightarrow[\text{converges rapidly as } n \rightarrow \infty]{} e^{-1} = 1/e \approx 0.368... \end{aligned}$$

Note: In fact, $d_n =$ closest integer to $\frac{n!}{e}$ $\forall n$.

Also have recurrence for d_n :

$$\text{Prop. } d_n = (n-1) \cdot (d_{n-1} + d_{n-2})$$

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